Pavel Tyapkin

Senior FPGA / RTL Design Engineer

Ph.D. equivalent (Candidate of Technical Sciences), Radio Engineering | Currently based in Latin America | Open to relocation / remote
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Professional Summary

FPGA/SoC engineer with 10 years of experience across satellite communications, navigation receivers, embedded platforms, and digital wireless systems. Specializes in **RTL development, FPGA integration, timing closure and resource optimization**, with hands-on experience bringing designs onto Xilinx 7-series hardware and connecting programmable logic to processors, memory and RF/data-acquisition peripherals.

Core Technical Skills

Area	Technologies and experience
RTL / FPGA	Verilog; Xilinx Vivado, ISE; Zynq-7000, Artix-7; Intel Quartus; module development, integration, migration between toolchain versions
Implementation / debug	Timing closure, RTL and resource optimization, hardware bring-up, ModelSim, Xilinx ILA/VIO, basic Verilator experience
FPGA interfaces	AXI/AXI-Lite, PS–PL DMA, PCIe Gen3 / XDMA, Ethernet, SPI, I²C, UART; ADC/DAC and RF peripheral interfaces
Embedded / SoC	Zynq PS/PL integration, memory-mapped registers, bare-metal-level peripheral checks, embedded Linux, Buildroot, C/C++, Python
DSP / communications	Digital filters, multi-channel acquisition and processing, OFDM, BPSK/QPSK, IEEE 802.11 PHY, MATLAB/Simulink, GNU Octave
SDR / laboratory	GNU Radio, USRP B210/N210, SDR-based measurement setups, RF signal generators, oscilloscopes
Engineering workflow	Git, code review, branch integration, technical task coordination; AI-assisted coding and research workflows (subject to project confidentiality requirements)

Professional Experience

Lead Radio Engineer — AO Energiya Razvitiya (ER)

October 2025 – October 2026

- Developed and integrated **FPGA RTL for IEEE 802.11-based wireless communication systems**, using established architectures as a starting point.
- Optimized designs for **timing closure and FPGA resource utilization**; maintained build compatibility through FPGA development toolchain updates.
- Adapted logic and integration flows to different FPGA/SoC hardware platforms and board configurations.

Technical details, implementation specifics and performance figures are omitted to respect confidentiality obligations.

FPGA / Embedded Systems Developer — NII KP, Branch of ORKK JSC

November 2020 – April 2025

- Designed and verified Verilog-based logic for **multichannel FPGA data acquisition and digital signal processing** on Xilinx 7-series FPGA/SoC platforms.
- Integrated data converters and peripherals, including ADC/DAC devices, memory and programmable RF components.
- Implemented and debugged **high-throughput data paths**, including PS–PL DMA and PCIe Gen3/XDMA, alongside SPI, I²C and UART control interfaces.

Engineer — Topcon Positioning Systems

September 2018 – September 2020

- Developed DSP algorithms and FPGA/DSP logic for navigation receiver equipment.
- Built SDR-based automated measurement and validation testbeds for radio hardware, including ETSI-related testing.
- Developed signal-generation and test automation software to support transmitter and receiver evaluation.

Engineering Technician — Russian Federal Space Agency

March 2016 – June 2018

- Participated in FPGA-based satellite communication system development and verification.
- Created simulation models to evaluate DSP processing stages and communication architectures.
- Developed hardware/software test setups for communication equipment.

Research & Education

Candidate of Technical Sciences (Ph.D. equivalent), Radio Engineering — Moscow Aviation Institute, 2025
Dissertation: *Improving the Interference Immunity of Radio Communication Systems Against Impulsive Interference Using Blind Source Separation Methods.*
Dissertation defense

M.Sc., Infocommunication Technologies and Communication Systems — Moscow Aviation Institute, 2020
B.Sc., Infocommunication Technologies and Communication Systems — Moscow Aviation Institute, 2018

Junior Researcher — Moscow Aviation Institute, 2023–December 2025
Studied ICA and blind source separation for interference mitigation in deep-space communications; designed experimental hardware prototypes and testbeds for algorithm evaluation as part of a Russian Science Foundation-funded project.

Teaching Instructor — Moscow Aviation Institute, 2018–June 2025
Led laboratory and tutorial sessions in Communication Theory and developed practical teaching materials.

Languages

- **Russian:** Native
- **English:** Advanced (C1, self-reported)